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Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources

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ABSTRACT Editing multi-channel video footage that is, producing a single refined output stream by selecting and switching among synchronized camera channels in conversations, discussions, and podcasts typically involves complex and labor-intensive processes conducted by human editors. Automating this task also poses significant challenges as it requires effectively capturing the nuanced decision-making processes inherent in professional editing. We propose a method to generate multi-hypothesis video edits that can be easily adapted to human editing intent from original multi-channel sources. Our method leverages advanced neural networks in both visual and auditory modalities to closely emulate the intricate decisions traditionally made by human editors. The proposed method generates multiple output hypotheses based on different modality configurations, providing human editors with valuable insights. In the visual modality, we first detect faces in each frame with a recognition model, and then estimate the speaking state for each detected region. In the auditory modality, speech recognition and speaker diarization are utilized to perform sentence-level segmentation and speaker identification. By synchronizing and integrating multimodal data, our approach automatically generates coherent and professionally edited video outputs. Experimental results demonstrate that our system achieves frame match accuracies of 84.99% and 90.10% for 3-channel and 5-channel sources, respectively. Furthermore, qualitative evaluations by experts in related fields indicate that our system is sufficiently suitable for practical applications.

INDEX TERMS Automated video editing, Multi-channel video editing, Multi-hypothesis video editing, Deep learning, Audio-visual processing

I. INTRODUCTION

An automated editing system aims to produce a refined video from multi-channel video footage, such as conversations, discussions, and podcasts. Traditionally, this has been a labor-intensive process, requiring human editors to carefully select and synchronize channels from multiple sources into a cohesive final stream. Developing an AI-based system to replicate this process is challenging, especially due to the lack of well-curated datasets that pair raw multi-channel footage with professional edits reflecting nuanced editorial decisions. Because such datasets are scarce, we do not rely on supervised learning from edited reference videos; instead, we replicate editor-like decisions by combining separate visual and auditory analysis pipelines and fusing their outputs, as described in Section III.

Recent video editing systems have attempted to automate specific tasks within the editing process. For instance, Koorathota et al. performed editing using visual and textual metadata [1]. However, their work relied on paired data in controlled environments, where sub-clips were annotated with text metadata, making it less suitable for general-purpose editing scenarios. Lomotin and Makarov proposed a method for evaluating frame quality in automated editing [2]. However, this approach was limited to visual analysis and did not account for audio cues or subjective evaluation factors, both of which are essential in practical applications.

Similarly, Wright et al. presented a rule-based AI system that performed shot framing, sequencing, and participant selection from wide-angle video streams [3]. While this method incorporated both visual and auditory information, it was designed specifically for structured formats such as live panel shows and did not address the challenges of multi-channel video editing. Additionally, while Adobe Premiere Pro's "Text-Based Editing" function enables rough cut generation via speech recognition and speaker diarization on master audio tracks, it lacks support for aligning transcripts to each channel video with individual input channels in multi-channel environments [4]. These studies demonstrated that existing systems were limited to solving individual sub-problems within the editing pipeline and that integrating both visual and auditory modalities for fully automated editing of multi-channel content remained to be explored.

To address these limitations, we propose an automated multi-channel video editing system that late-fuses multiple modalities through distinct pipelines for visual and auditory analysis. In the visual stream, we perform frame-level analysis of each channel by leveraging deep models to detect faces and estimate their speaking state. In the auditory stream, we employ speech recognition and speaker diarization models to perform sentence-level segmentation and speaker identification. The outputs of these two pipelines are then synchronized and integrated to generate coherent, professional-quality editing results from multi-channel video footage.

Considering both visual and auditory modalities is crucial for effective multi-channel editing. A visual-only approach can be ambiguous when multiple faces are visible, when the active speaker is partly occluded, or when speaking-state estimates are noisy; in such cases, audio-based speaker diarization provides a clear "who is speaking" signal that disambiguates which channel to display. Conversely, audio alone does not indicate which camera channel corresponds to each speaker; pairing diarization with per-channel visual speaking-state estimates allows the system to map speakers to channels and select the correct shot. This multimodal integration yields substantial gains over a unimodal baseline: as shown later in Table 1, combining visual and speech (Visual+Speech) achieves markedly higher frame-level match rates than using visual information alone (e.g., from 79.18% to 84.60% for 3-channel and from 73.53% to 89.77% for 5-channel video without beam search), illustrating why both modalities are necessary for robust editing.

The main contributions of our paper are as follows:

- An automated system for editing multi-channel video by synchronizing raw inputs through separate visual and auditory pipelines.
- Deep late-fusion of multimodal information for face recognition, speaking-state estimation, speech recognition and speaker identification.
- A multi-hypothesis editing approach that provides candidate versions based on probability distribution over time for extracting frames from each channel stream.
- Frame-level match accuracies of 84.99% and 90.10% for the 3-channel and 5-channel videos, highlighthing the system's robust performance across varying speaker complexities.

Our approach differs from data-driven methods that require large datasets of professionally edited references: we combine off-the-shelf ASR and diarization with a learned visual speaking-state model and beam search, avoiding the need for paired edit labels. It also differs from single-camera or rule-based systems [3]: we address synchronized multi-channel studio footage and provide multi-hypothesis outputs that support editor-in-the-loop workflows.

II. RELATED WORK

Recent advancements in automated multi-channel video editing have increasingly leveraged deep neural networks to enhance multimodal analysis and editing processes. Key research areas include score-based editing, which utilizes quantitative metrics to select optimal shots for improved video coherence and quality. Selecting shots using virtual view, adjust shot levels and angles based on objects or performers of interest. And Audio based editing, which analyzes and transcribes audio to detect events or speech segments that inform editing decisions. Additionally, significant progress in automatic speech recognition, speaker diarization, and object detection and image classification has contributed to more accurate and efficient analysis, has enhanced automated editing capabilities.

Score-Based Editing focuses on enhancing the editing process by evaluating and selecting shots or sequences based on quantitative metrics or predefined quality scores. Several studies have proposed to provide frameworks and methods to automate video editing through score-driven evaluation. For instance, Podlesnyy proposes a data-driven approach for automatic video editing, where editing decisions are guided by metrics learned from datasets [5]. This study presents the importance of understanding human aesthetic preferences and embedding them into machine learning models to improve the quality and coherence of edited videos. Similarly, Lomotin and Makarov introduced an automated image and video quality assessment tailored for computational video editing [2]. By utilizing perceptual quality metrics, this approach identifies high-quality shots and filters out undesirable footage, ensuring a more polished and professional outcome in the automated editing pipeline. Furthermore, Huang*et al.* presents SB-VQA (Stack-Based Video Quality Assessment), a framework designed to assess video quality using a stack-based approach [6]. The method provides accurate quality scores for video sequences, to support video enhancement tasks and ensures that the selected shots meet predefined quality thresholds for effective editing. This approach demonstrates strong potential in optimizing editing workflows, particularly in scenarios where maintaining high video quality is critical.

Selecting Shots Using Virtual Shot is an automated editing method that adjusts the level and angle of shots based on performers or objects of interest within a single wide-angle recorded video. For example, Moorthy et al. and Achary et al. propose systems that automatically generate cinematic edited videos from footage captured by a single static wide-angle camera, utilizing virtual camera techniques and gaze data [7], [8]. These methods are structured into two main stages: virtual camera generation and shot selection. Additionally, Wright *et al.* performs shot sequencing using face detection and tracking, facial landmarking, visual speaker detection, and audio event detection [3]. The system evaluates a shot levels and selection based on the number of detected individuals and an analysis of the speaker's voice volume.

Using automatic speech recognition (ASR) makes it possible to convert spoken language into text, serving as a foundational component in the analysis of dialogue within multi-channel video content. While early models set the foundation, the practical utility of modern ASR is defined by its application in real-world systems. One of the primary use cases is automated subtitling, which is critical for enhancing content accessibility across various domains. Recent studies, such as those conducted by Rao [9], have demonstrated the effectiveness of robust ASR models like Whisper in transcribing educational videos. These results highlight the potential of modern ASR to substantially reduce both the cost and time associated with manual transcription. Moreover, ASR now functions as a critical preprocessing step in more complex video analysis pipelines. For example, transcripts generated by ASR can be further processed using Large Language Models (LLMs) to automatically generate video summaries [10]. To address the persistent challenges posed by background noise, research has increasingly focused on multimodal approaches. Audio-Visual Speech Recognition (AVSR) systems, such as Whisper-Flamingo, integrate visual modalities with audio input. This multimodal integration has significantly improved transcription accuracy under noisy conditions, outperforming traditional audio-only ASR models [11]. Recently, Whisper has been extended to jointly perform transcription, alignment, and diarization in a single model (Whisper-TAD), illustrating the trend toward integrated speech processing that our pipeline leverages via ASR and diarization components [12].

Using speaker diarization is not a standalone analysis for videos but becomes very useful when integrated with other speech processing techniques [13]. A prominent application of speaker diarization is its combination with ASR to generate speaker-attributed transcripts. Open-source toolkits such as pyannote.audio provide pre-trained neural network models and then facilitates the deployment of this technology in practical settings [14]. This integration is particularly valuable in multi-speaker environments and is critical as the transcription itself in business meetings or clinical consultations [15], [16]. To handle complex scenarios with overlapping speech, research has focused on End-to-End Neural Diarization (EEND) models. These models jointly optimize speaker segmentation and identification within a unified architecture and have been adapted for streaming scenarios to support real-time processing [17], [18]. Word-level end-to-end diarization (WEEND) has further combined ASR and diarization to assign speaker labels at the word level, improving alignment for downstream tasks [19]. Audio-Visual Speaker Diarization (AVSD) systems has incorporated visual information, such as lip movements and facial identity, to resolve speaker ambiguities that cannot be distinguished through audio alone [20].

Visual analysis-based editing now plays a direct role in automated decision-making within video editing pipelines. A core component is Active Speaker Detection (ASD), which identifies the speaking individual in a video to inform camera focus. This is typically achieved by integrating real-time face detection such as YOLO with synchronized audio cues [21]. Recent work has pushed ASD toward long- and short-term context modeling: for instance, LoCoNet achieves state-of-the-art results on multi-speaker benchmarks by combining intra-speaker temporal modeling with inter-speaker interaction modeling [22]. To support real-time, on-device deployment, other work has focused on developing lightweight ASD models that balance high accuracy with low computational overhead, enabling seamless integration into editing software [23]. In parallel, transformer-based architectures such as the Swin Transformer have emerged as versatile backbones for diverse video analysis tasks, including action recognition, and have been incorporated into detection frameworks to enhance performance [24], [25]. These advancements converge in systems like EditIQ, which automates cinematic editing by combining ASD, dialogue understanding via LLMs, and visual saliency estimation. EditIQ formulates editing as an optimization problem, selecting shots that balance narrative importance with cinematic coherence and aesthetic quality [26].

III. PROPOSED METHOD

For multi-speaker, multi-channel content, we adopted an intuitive approach to make automated editing feasible. This is primarily because datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare, as they would need to represent highly specific editing scenarios. The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automated video editing.

To overcome this limitation, we focused on replicating the editing actions of human editors through artificial neural networks for each modality. Rather than relying on curated datasets of edited videos, we aimed to substitute the decision-making processes of human editors with deep learning models capable of processing audio, visual, and other relevant data independently.

In our approach, ASR was employed to retrieve spoken content from audio channels, and Speaker Diarization was utilized to identify who was speaking at each point in time. Speaking State Estimation helped determine when individuals were actively contributing to the discussion. Visual features, including face detection and facial expression analysis, were also leveraged to provide contextually appropriate edits. Each neural network model was trained to handle a specific aspect of the editing process, allowing us to create a system capable of understanding the dynamics of multi-speaker interactions. By addressing each modality audio, visual, and derived outcome independently and then integrating these insights, our proposed system can adapt to diverse editing scenarios without relying on pre-labeled datasets. This approach, integrating multiple neural networks, ultimately enabled our system to replicate the nuanced and context-aware decisions typical of human editors, thereby reducing manual intervention while preserving the quality and coherence of the edited video content.

A. AUDIO MODALITY

For the audio modality, our goal is to accurately detect active speech segments and identify the corresponding speakers from diverse and often noisy multi-channel sources. To achieve this, we selected a large-scale, weakly supervised ASR model[1] known for its robustness across various languages and accents. For the precise temporal alignment crucial for video editing, we then integrated an open-source ASR extension[2] that provides sentence-level timestamps.

However, the sentence-level timestamps from the ASR tool can be imprecise. To resolve this limitation and assign speaker labels, we employ an open-source speaker diarization toolkit[3] to perform speaker diarization. The process involves first using the ASR tool to generate an initial transcription for a sentence S with a coarse time interval $T = (t_{\text{start}}, t_{\text{end}})$. Subsequently, the speaker diarization toolkit produces precise speaker segments, each with a speaker identity and temporal boundaries. To assign a single, dominant speaker to the entire sentence S, we identify all speaker segments from the diarization output that temporally overlap with the sentence's interval T. For each unique speaker found within this interval, we calculate their total speaking duration. The speaker with the highest cumulative speaking duration within the sentence's timeframe is then assigned to that entire sentence. This sentence-level assignment strategy ensures that each transcribed sentence is attributed to a single speaker, which is essential for the subsequent editing process.

B. VISUAL MODALITY

The visual approach involves estimating the speaking state based on the facial region of the speaker. This method requires accurate detection of the face region, which serves as the basis for analyzing facial expressions and movements related to speaking. To achieve this, we perform face region detection for each frame, allowing us to track and analyze even subtle facial dynamics in real time.

To address scenarios where the speaker's face is partially or fully obstructed or where sudden video changes occur, we incorporate an error-handling mechanism. Specifically, when the face detection model fails to locate a face, the system interpolates the bounding box B using temporal smoothing methods based on previously detected frames. Additionally, a confidence score threshold is applied to the output of the detection model, ensuring only reliable face detections are utilized.

We employ a robust and efficient face detection toolkit[4] that provides a lightweight model to identify the bounding box of the face within each frame, represented as $B = (x_{\text{min}}, y_{\text{min}}, x_{\text{max}}, y_{\text{max}})$. This enables us to extract the face region of interest (ROI), which is crucial for further analysis.

Using this detected face ROI, we estimate the speaking state using a vision transformer architecture[5] configured for binary classification. The extracted face ROI is fed into the vision transformer, which predicts whether the speaker is in an active speaking state. The model uses a final classification head with a sigmoid activation function to produce a probability value, P_{speaking}, between 0 and 1. The classification process is defined as:

$$P_{\text{speaking}} = \sigma(W_{\text{fc}} F + b_{\text{fc}})$$

where W_{fc} and b_{fc} are the weights and bias of the fully connected layer, F represents the features extracted by the vision transformer, and σ is the sigmoid function. The final decision rule is: the frame is classified as speaking if $P_{\text{speaking}} > 0.5$, and as non-speaking otherwise. This architecture benefits from the vision transformer's ability to efficiently handle visual features and capture both local and global context within the face ROI, thus significantly improving the accuracy of speaking state detection.

C. INTEGRATION AND CHANNEL SELECTION

In multi-channel systems, accurately aligning speaker activity with the most appropriate video channel is essential. Our system achieves this by combining frame-level visual speaking likelihoods with temporal speaker activity segments derived from audio analysis.

The process begins by determining, for each frame, which channel most likely corresponds to an active speaker by selecting the one with the highest predicted speaking probability. In parallel, speaker activity segments from the audio modality are converted into frame indices, assigning a speaker label to each video frame. To account for diarization noise or short interruptions, we apply a smoothing technique detailed in Algorithm 3. This method merges transient segments shorter than a predefined duration (e.g., 30 frames) with adjacent speaker labels, ensuring visual stability and reducing unwanted jitter in the final edit.

[1] OpenAI's *Whisper* [27].
[2] Pretrained *WhisperX* [28].
[3] *Pyannote.audio* [29], [30].
[4] MediaPipe's *Face Detection* [31], [32].
[5] *Swin Transformer* [33].

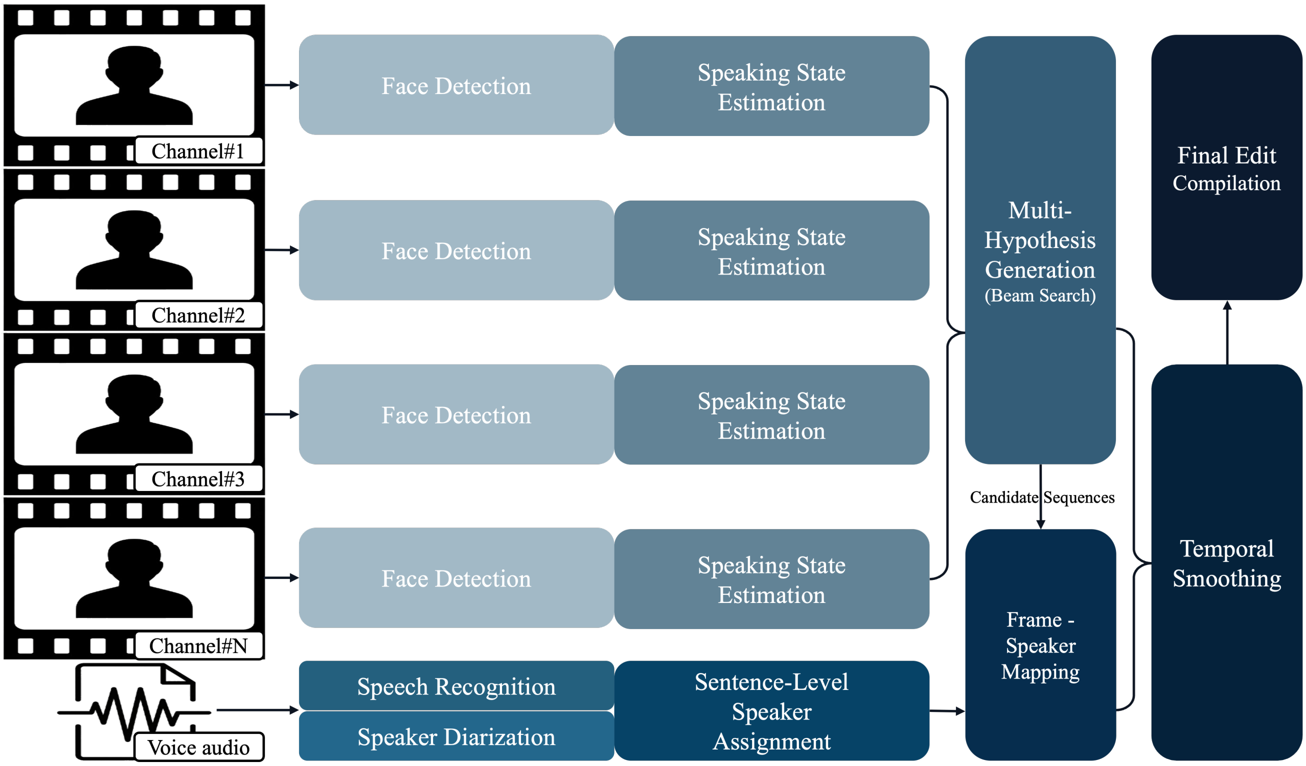


FIGURE 1: The overall architecture of our proposed system. Multi-channel video and audio inputs are processed through two parallel pipelines. The visual pipeline (top) performs face detection and speaking state estimation per channel, yielding per-frame face bounding boxes and speaking probabilities per channel. The audio pipeline (bottom) uses speech recognition and speaker diarization to produce sentence-level segments with speaker IDs and timestamps. These outputs are integrated: the Multi-Hypothesis Generation module (Beam Search) takes per-frame channel likelihoods and transition costs to output a sequence of channel indices over time. A Temporal Smoothing module then refines short spurious segments before the final edited video is compiled.

Once both frame-wise speaker labels and channel predictions are established, we associate each speaker with the channel that most frequently coincides with their speaking intervals. This yields a primary speaker-to-channel mapping. However, to create a truly coherent final output, we refine this mapping using a more robust hypothesis generation method.

Multi-Hypothesis Generation for Temporal Coherence

A naive frame-by-frame selection of the best channel can lead to frequent and abrupt transitions, disrupting the viewing experience. To generate temporally coherent editing sequences, we formulate the task as finding an optimal channel sequence $C = (c_1, \ldots, c_T)$ that maximizes a joint log-probability score over the entire video. Let $f_{1:T}$ denote the sequence of visual observations over T video frames, where f_t corresponds to the visual features extracted at frame index t. Our goal is to infer the most likely sequence of channel assignments $c_{1:T} = \{c_1, c_2, \ldots, c_T\}$, where $c_t \in C$, the set of available channels. Thus, we formulate the inference problem as a maximum a posteriori (MAP) estimation:

$$\hat{c}_{1:T} = \underset{c_{1:T}}{\text{argmax}} P(c_{1:T}|f_{1:T}) \tag{1}$$

Assuming a first-order Markov dependency, the joint probability can be factorized as:

$$P(c_{1:T}|f_{1:T}) \propto P(c_1|f_1) \prod_{t=2}^{T} P(c_t|f_t) \cdot P_{\text{transition}}(c_{t-1}, c_t) \tag{2}$$

where $P(c_t|f_t)$ denotes a channel likelihood at frame t, given the visual observation f_t, and $P_{\text{transition}}(c_{t-1}, c_t)$ means a channel transition probability between consecutive frames.

This can be thought of as a score function that balances the likelihood of a channel being correct at a given frame with the cost of transitioning between channels. The transition probability is defined to favor continuity in channel selection to improve temporal smoothness.

$$P_{\text{transition}}(c_{t-1}, c_t) = \begin{cases} 0.9, & \text{if } c_{t-1} = c_t \\ \frac{0.1}{|C|-1}, & \text{otherwise} \end{cases} \tag{3}$$

where, $|C|$ represents the total number of available channels. This probabilistic framework ensures that the selected channel sequence maintains temporal smoothness, only switching channels when the visual evidence f_t strongly supports a change. Since finding the exact sequence that maximizes

Equation (2) is computationally expensive, we employ beam search to effectively find the optimal sequence.

As detailed in Algorithm 1, this method approximates the optimal solution by iteratively extending the top-k most probable channel sequences (the "beam") at each frame. A key aspect of this algorithm is its handling of ambiguity: if the confidence in a single speaker is low (i.e., the difference between the highest and second-highest speaker probabilities is below a 'threshold'), the system defaults to a "wide channel" shot to ensure stability. This multi-stage process, combining Beam Search (Algorithm 1) with a subsequent unique channel assignment based on occurrence frequency (Algorithm 2), allows us to generate multiple high-quality, temporally consistent hypotheses for the final edit. This significantly enhances robustness and flexibility in dynamic, multi-speaker video editing workflows.

Algorithm 1: Frame-wise Log-Probability Construction and Beam Search Decoding

Input: speaking probabilities for all speakers per frame, total_frames, transition_log_prob, beam_width (default: 3), threshold
Output: Top-k decoded channel paths

Step 1: Frame-wise Log-Probability Construction
for $t \leftarrow 0$ **to** $total\_frames - 1$ **do**
    foreach *speaker s* **do**
        $p_s \leftarrow$ speaking probability at frame t;
    end
    if *fewer than 2 valid speakers* ***or*** $(\max(p) - \textit{second-max}(p)) < threshold$ **then**
        Assign full probability to *wide channel*, set all $p_s \leftarrow 0$;
    else
        Normalize p_s over all speakers;
        Set $p_{\text{wide}} \leftarrow 0$;
    end
    Convert all p to log-probabilities and store in $frame\_probs[t]$;
end
Step 2: Beam Search Decoding
Initialize $beams \leftarrow$ list of $(empty\ path, 0.0)$;
foreach t *in* $frame\_probs$ **do**
    $new\_beams \leftarrow$ empty list;
    foreach $(path, score)$ *in beams* **do**
        $prev \leftarrow$ last channel in $path$ (or None);
        foreach $(channel, \log p)$ *in* $frame\_probs[t]$ **do**
            $trans \leftarrow transition\_log\_prob[(prev, channel)]$ or 0;
            $new\_score \leftarrow score + trans + \log p$;
            Append $(path + [channel], new\_score)$ to new_beams;
        end
    end
    Keep top-k elements in new_beams by descending score;
    $beams \leftarrow new\_beams$;
end
return *Top-k paths in beams sorted by score*

IV. EXPERIMENT

A. EXPERIMENTAL SETUP

1) Audio Modality

For processing the audio modality, we employ a pre-trained automatic speech recognition model, whisper-large-v2, along with a pre-trained speaker diarization framework, Pyannote.audio. These components are integrated using WhisperX, enabling robust extraction of linguistic features, precise temporal alignment at the word and sentence levels, and reliable speaker identification.

2) Visual Modality

For processing the visual modality, we employ Face Detection to extract Face ROIs. To estimate speaking states, we train the Swin transformer model using the dataset described below.

- Dataset: We utilized the Lip Reading dataset [34], which comprises speech videos recorded with various speakers and filming angles. We selected frames corresponding to utterance times specified in the labels, as well as those outside these intervals, and applied Face Detection to identify and extract Face ROIs. Each Face ROI was assigned a binary label indicating whether it represented an active utterance or silence. This process yielded a large-scale dataset to facilitate robust model training and evaluation. The training set contained approximately 7,975,028 utter face images and 6,740,797 silence face images, while the validation set consisted of about 872,377 utter face images and 526,647 silence face images.
- Training Details: We trained and validated the Swin transformer on the previously described dataset for 10 epochs, using a learning rate of 5e-4 and the Adam optimizer, in conjunction with a Noam learning rate schedule to ensure stable convergence. The training process was distributed across six A6000 GPUs with PyTorch's DataParallel framework. As a result, the model achieved a validation accuracy of 95.88%.

3) Evaluation Data Description

To evaluate our system, we utilize two multi-channel video datasets from studio recordings provided by a Korean public broadcasting company in South Korea, consisting of a two-speaker interview and a four-speaker panel discussion. In both datasets, the raw, synchronized recordings from every camera channel are available.

- **3-channel video:** An interview involving two speakers. Frame-level speaker labels are provided for each channel, but there is no professionally edited reference video.

Algorithm 2: Unique Channel Assignment per Speaker

```
Input: multi_channel_data (beam path),
       transcription_data, fps (default: 30)
Output: speaker_channel_mapping
Compute frame indices per speaker segment;
Collect channel occurrences per speaker;
while unassigned speakers remain do
    Select highest occurrence speaker-channel pair;
    Assign channel uniquely to speaker;
    Remove assigned channel from other speakers;
end
return speaker_channel_mapping;
```

Algorithm 3: Algorithm for Smoothing Short Segments in Speaker Data

```
Input: speaker_data (list of speakers), threshold
       (default: 30)
Output: smoothed_speaker_data (list of speakers)
Convert speaker_data to an array, n ← length of
speaker_data, segment_start ← 0;
for i ← 1 to n do
    if i = n or speaker_data[i] ≠
      speaker_data[segment_start] then
        if i - segment_start ≤ threshold then
            left ← speaker_data[segment_start - 1] if
              segment_start > 0 else None;
            right ← speaker_data[i] if i < n else None;
            speaker_data[segment_start : i] ← left if
              left = right else (left or right);
        end
        segment_start ← i;
    end
end
return speaker_data as a list;
```

- **5-channel video:** A panel discussion with four speakers. This dataset includes both frame-level speaker labels and a professionally edited reference video created by a human editor.

Both datasets are additionally divided into sub-intervals that coincide with the editorial decision units used by human editors.

B. EVALUATION

We compared the ground-truth labeled data with the output generated by our system at the frame level. By quantifying the match rate under various criteria and providing corresponding visualizations, we assessed the overall effectiveness and robustness of our system.

It should be noted that the Wide channel, which is selected when there is insufficient evidence for channel selection or ambiguous conditions, is not used for analysis but is employed only in the final channel selection process.

			Visual+Speech + Beam		
Data	Visual	Visual+Speech	Thr 15	Thr 30	Thr 45
No beam					
3-channel video	0.7918	0.8460	–	–	–
5-channel video	0.7353	0.8977	–	–	–
Beam (bias 0.7)					
3-channel video	–	–	0.8498	0.8495	0.8495
5-channel video	–	–	0.8995	0.8999	0.8854
Beam (bias 0.9)					
3-channel video	–	–	**0.8499**	0.8497	0.8481
5-channel video	–	–	0.8987	**0.9010**	0.8865

TABLE 1: Match rates: Baseline (Visual/Visual+Speech) and Beam Search (varying threshold).

Length (s)	**3-channel video**			**5-channel video**		
	Match	Frames	Rate	Match	Frames	Rate
0–5	3,849	6,660	0.5779	2,981	4,335	0.6877
5–10	4,162	4,695	0.8865	5,447	5,940	0.9170
10–20	5,161	5,340	0.9665	2,867	4,425	0.6479
20–30	2,971	3,210	0.9255	3,573	3,720	0.9605
30–40	1,140	1,140	1.0000	8,412	8,580	0.9804
40–50	–	–	0.0000	2,524	2,550	0.9898
50–60	3,104	3,315	0.9363	1,648	1,650	0.9988
60–70	–	–	0.0000	2,010	2,010	1.0000
70–80	2,128	2,130	0.9991	2,198	2,220	0.9901
80–90	–	–	0.0000	2,670	2,670	1.0000
Total	22,515	26,490	**0.8499**	34,330	38,100	**0.9010**

TABLE 2: Match rates across different segment lengths.

V. RESULTS

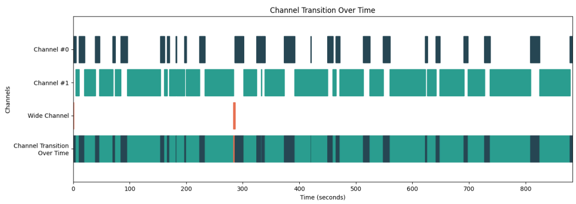
In this section, we present both quantitative and qualitative evaluations of our proposed multi-camera video editing system. We first analyze the frame-level alignment accuracy under various smoothing and temporal conditions, followed by subjective assessments of editing quality and user experience.

A. QUANTITATIVE ANALYSIS

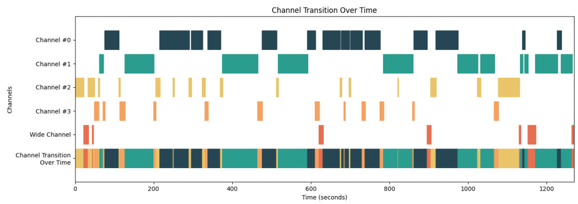
We quantitatively evaluated the performance of our system by analyzing frame-level match rates under several conditions: the benefit of multimodal integration and the effect of self-transition bias and smoothing threshold (Table 1), and the effect of speech segment length (Table 2).

1) Benefit of Multimodal (Visual+Speech) Integration

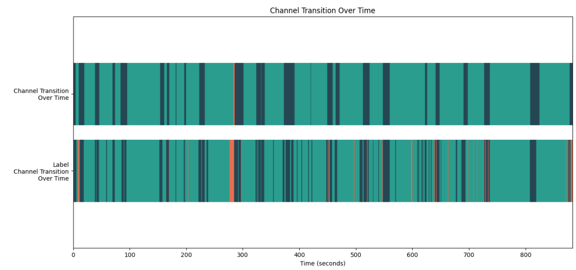
To demonstrate why integrating both visual and auditory modalities is essential, we compare a visual-only baseline with our full Visual+Speech pipeline (no beam search). Table 1 reports the frame-level match rates: the “No beam” rows give Visual and Visual+Speech baselines. For the 3-channel video, adding speech improves the match rate from 79.18% (visual only) to 84.60% (Visual+Speech); for the 5-channel video, the gain is larger, from 73.53% to 89.77%. In the 5-channel setting, multiple speakers and channels make visual-



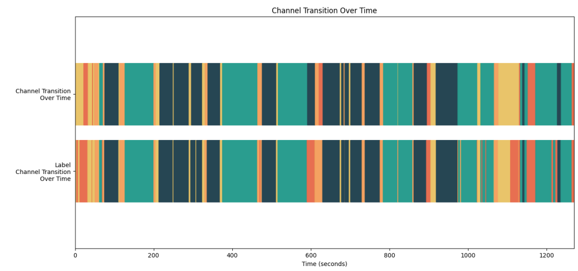
(a) Channel transition over time (3-channel).



(c) Channel transition over time (5-channel).



(b) Transition and labels across segments (3-channel).



(d) Transition and labels across segments (5-channel).

FIGURE 2: Visualization of channel selection for the 3-channel (left column) and 5-channel (right column) videos. Each sub-figure shows the overall transitions and a detailed view with ground-truth labels.

only selection more error-prone, whereas speaker diarization from audio provides a stable "who is speaking" signal and thus a stronger improvement. The same table also shows that applying beam search on top of Visual+Speech yields the best overall rates (84.99% and 90.10%). These results support the argument in the introduction that multimodal integration offers significant advantages over a unimodal (visual-only) approach.

2) Effect of Smoothing Threshold

The Visual+Speech + Beam section of Table 1 shows the system's performance when applying different smoothing thresholds (15, 30, and 45 frames) under two self-transition bias settings (0.7 and 0.9). For the 3-channel video, the best match rate was achieved with a threshold of 15 frames under a bias of 0.9 (0.8499), although the difference across thresholds was minimal, indicating relatively stable performance. In the 5-channel video, the highest match rate (0.9010) was obtained at 30 frames with a self-transition bias of 0.9, suggesting that moderate smoothing helps prevent overfitting to transient speaker fluctuations. These results highlight that an intermediate smoothing threshold of 30 frames generally provides optimal performance, balancing responsiveness to genuine speaker transitions and robustness against noise.

3) Performance by Segment Length

Table 2 details match rates across varying speech segment durations for both 3-channel and 5-channel videos. The evaluation reveals a strong correlation between segment length and match accuracy. For the 3-channel video, shorter segments such as 0–5 seconds yielded lower accuracy (0.5779), while longer segments such as 30–40s and 70–80s achieved near-perfect match rates (1.0000 and 0.9991, respectively). Notably, several duration intervals (e.g., 40–50s, 60–70s, 80–90s) contained no speech segments and were excluded from aggregate scoring. In contrast, the 5-channel video demonstrated high accuracy across nearly all segment durations. Even shorter segments (0–5s) achieved a match rate of 0.6877, and the performance steadily improved with longer durations, reaching 1.0000 in three intervals (60–70s, 80–90s, and 40–50s) and exceeding 0.98 in others.

4) Overall Accuracy

Across all evaluated frames, the system achieved a total match rate of 0.8499 for the 3-channel video (22,515 matched frames out of 26,490) and 0.9010 for the 5-channel video (34,330 matched out of 38,100). These results demonstrate the system's robustness and generalizability, particularly in handling complex multi-speaker scenarios. Table I also serves as an internal comparison: the Visual-only row is representative of prior visual-centric or single-camera approaches, and the gains from Visual+Speech and from beam search quantify the benefit of our full multimodal pipeline. In summary, the proposed system benefits from both moderate smoothing thresholds and longer speech segments, and maintains consistent high accuracy even in challenging multi-speaker environments.

B. QUALITATIVE ANALYSIS

To assess the perceptual quality of our system's output on the aforementioned datasets, we conducted a subjective evaluation. The analysis was performed with 24 participants, including developers and broadcasting industry professionals, under a blind setup, where they were not informed whether a

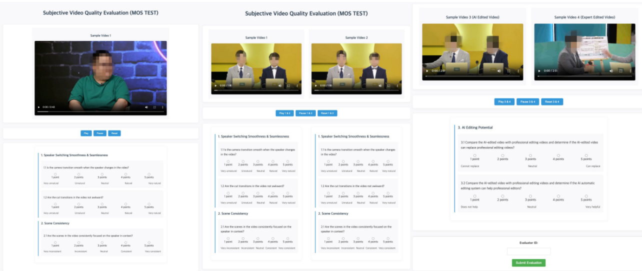


FIGURE 3: Subjective Video Quality Evaluation (MOS Test) Interface.

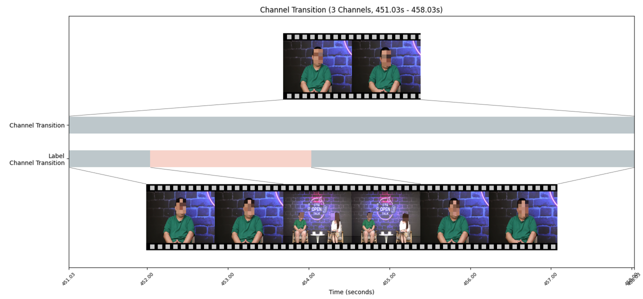
Criteria	3-channel Ours	5-channel Ours	5-channel Expert
1. Speaker Switching Smoothness & Seamlessness			
1.1 Is the camera transition smooth when the speaker changes in the video?	4.26±0.81	4.43±0.78	3.95±1.02
1.2 Are the cut transitions in the video not awkward?	4.26±1.00	4.34±0.64	4.04±1.02
2. Scene Consistency			
2.1 Are the scenes in the video consistently focused on the speaker in context?	4.17±0.88	4.52±0.59	3.95±0.92
3. AI Editing Potential			
Compare the AI-edited video with professional editing videos and determine			
3.1 if the AI-edited video can replace professional editing videos?	–	4.08±1.08	
3.2 if the AI automatic editing system can help professional editors?	–	4.56±0.66	

TABLE 3: Subjective Video Quality Evaluation Results.

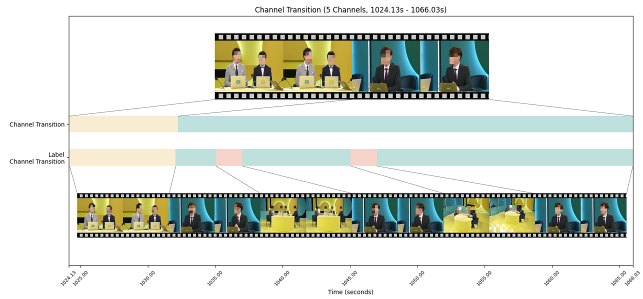
video was edited by our system or by a human expert. This blinding ensures that ratings are based solely on perceptual quality, mitigating any bias related to the editing source. The detailed evaluation results are summarized in Table 3.

1) Speaker Switching Smoothness and Seamlessness

As shown in Table 3, our system received high ratings for smooth transitions during speaker changes. Specifically, the edited videos obtained by our system achieved average scores of 4.26 for 3-channel videos and 4.43 for 5-channel videos, both exceeding the score of 3.95 obtained by professionally edited video. To ensure an unbiased evaluation, we adopted a double-blind setup in which participants were not informed whether a given video was edited by our system or by a professional human editor. This design allowed the assessment to focus purely on the perceptual quality of speaker transitions, eliminating potential biases. These results demonstrate that our system can generate visually natural and seamless speaker transitions. In some cases, it even outperforms traditional manual editing in terms of perceived visual continuity.



(a) 3-channel video transition preview.



(b) 5-channel video transition preview.

FIGURE 4: Comparison of our system's edits to ground-truth labels. (a) 3-channel: The system maintains speaker focus while the label indicates a wide shot. (b) 5-channel: The system creates a more stable edit by avoiding the brief wide-shot cuts present in the labels.

2) Scene Consistency

Our system also exhibited strong performance in maintaining scene consistency, which we define as the visual alignment between the displayed speaker and the actual active speaker. In terms of subjective evaluation, our system achieved average scores of 4.17 for 3-channel videos and 4.52 for 5-channel videos, both outperforming the expert-edited video, which scored 3.95. This improvement in scene consistency can be partially attributed to our system's ability to select contextually appropriate wide-angle shots. For example, as illustrated in Fig. 4, during specific segments where the ground-truth annotations indicated that a wide view was appropriate even while a speaker was active (e.g., 451.30s 458.03s in the 3-channel video and 1024.13s 1066.03s in the 5-channel video), our system's choices were rated favorably. These results suggest that the integration of audio-based speaker diarization with vision-based active speaker detection enables our system to maintain a consistent visual focus on the correct speaker more effectively than conventional manual editing approaches.

3) AI Editing Potential

The potential of our system to replace or assist professional editing was assessed using the 5-channel video scenario. As presented in the lower section of Table 3, the system-edited video received a score of 4.08 for its ability to serve as a full replacement for professional editing, and a notably higher score of 4.56 for its usefulness in supporting professional editors. These results suggest that while complete editing automation may not yet be universally viable, the system demonstrates significant potential as a productivity-enhancing tool. The proposed system can reduce manual effort and help decision-making without compromising perceptual quality. Therefore, it can serve as an effective augmentation to human expertise in real-world video production environments.

VI. DISCUSSION

We do not claim a fundamental algorithmic breakthrough; our contribution is a practical system that integrates existing and learned components for multi-channel studio editing, a setting where end-to-end learned solutions are currently infeasible due to data scarcity. The effectiveness of our system is attributed to its hybrid architecture: the ASR and speaker diarization models provide precise audio-based speaker segmentation, while the vision transformer-based visual modules enhance temporal alignment by detecting speaking activity. In ambiguous scenarios, a fallback to the wide-angle view ensures continuity, although such frames were excluded from this analysis. Overall, the system consistently generated coherent, contextually relevant edits and demonstrated practical utility in multi-speaker video production.

Comparison with prior work. Direct numerical comparison with existing automated editing methods is limited because our task frame-level channel selection from synchronized multi-channel studio footage, evaluated by match rate against reference channel labels is not addressed under the same setting or metric in prior work. Wright *et al.* [3] operate on single wide-angle streams with rule-based shot selection; EditIQ [26] produces virtual camera feeds from a single static view and uses different evaluation protocols. We therefore provide internal comparison through ablations in Table I: the Visual-only baseline is representative of approaches that rely solely on speaking-state or similar visual cues (as in many single-camera or visual-centric systems), and the gains from adding speech (Visual+Speech) and from beam search demonstrate the contribution of each component. The qualitative comparison with expert-edited video (Table III) and the expert assessment of our outputs further indicate that the proposed system reaches a level suitable for practical use. We view the combination of these ablations and expert evaluation as the appropriate form of comparison for this task until shared benchmarks or reimplementable baselines for multi-channel frame-level editing become available.

Future work could extend this framework in several directions. First, incorporating more sophisticated directorial logic, such as analyzing listener reactions or group dynamics, could enable more context-aware editing choices beyond focusing solely on the active speaker. Second, exploring end-to-end trainable models that jointly optimize all modalities could further streamline the system and potentially uncover more complex editing patterns. Finally, expanding the evaluation to a wider variety of content genres and languages would further validate the generalizability of our approach.

VII. CONCLUSION

In this paper, we introduced an automated multi-channel video editing system designed to replicate the nuanced decision-making of human editors. Our approach successfully addresses the key challenge of data scarcity by integrating distinct audio and visual processing pipelines, rather than relying on supervised learning with paired edited footage. By leveraging deep models for audio analysis (speech recognition and speaker diarization) and visual analysis (face detection and speaking state estimation), our system accurately estimates speaker activity and speaking states. These multimodal insights are then fused using a multi-hypothesis Beam Search approach to generate coherent, professional-quality video edits.

Our extensive experiments demonstrate the effectiveness of this system. Quantitatively, it achieved high frame-level match accuracies of 84.99% and 90.10% on 3-channel and 5-channel datasets, respectively. Qualitatively, in blind evaluations, our system's edits demonstrated highly competitive performance, showing results comparable to those produced by a human expert, particularly in transition smoothness and scene consistency. Furthermore, evaluators confirmed the system's significant potential as a tool to assist professional editors.

The primary contribution of this work is the demonstration that a multimodal system can effectively automate complex video editing tasks in real-world scenarios. By breaking down the editing process into specialized sub-tasks handled by expert models, we created a robust and adaptable framework. Future work will focus on incorporating more sophisticated directorial logic and expanding the system's evaluation to a wider variety of content.

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